

Global Facilities - Design & Construction Standard – CSA

- Concrete Works for Precast Specification

ECN Facility:	GLOBAL FACILITIES	ECN Area:	Engineering
Owner(s)	Engineering	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	
Version:	1	A3YRXSD74VDV-57553043-539	
Current ECN:	101127004	Last Modified:	09/22/2022

[ECN History](#)

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1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Concrete Works for Precast Specifications as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.
- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for concrete works for precast specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFCT	Micron's Global Facilities Construction Team
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 Material

4.1 General

In this Specification, the term 'Contractor' shall apply to Manufacturer / Precaster of precast concrete elements / components.

Materials used in the works shall be of the qualities and kinds specified herein. No materials shall be used until and unless approved.

Materials shall be transported, handled and stored on the site or elsewhere in such a manner as to prevent damage, deterioration or contamination all to the satisfaction of the Engineer.

Contractor shall submit to the Engineer the appropriate certificate of compliance with the Standards. Notwithstanding the above, tests shall be carried out as and when directed by the Engineer in accordance with the relevant Standards.

4.2 Cement

Unless otherwise specified or instructed by the Engineer, cement shall be of an approved brand and manufacture and comply with the following standards or equivalent local codes:

<u>Type</u>	<u>Relevant Standard</u>
Ordinary Portland Cement	SS EN SS 197-1
Rapid Hardening Portland Cement	SS EN SS 197-1

High alumina cement shall not be used for any concrete work.

Before starting any work the Contractor shall submit for approval the type and brand of cement he proposes to use. The Contractor shall not change the cement brand unless he has specific reasons to do so and permission in writing shall be obtained from the Engineer.

Contractor shall produce a certificate showing the date and place of manufacture of the cement. As and when directed by the Engineer, samples of cement shall be provided for testing. Any cement failing to comply with the requirements of the relevant standard shall be rejected.

The manufacturer's test certificate will normally be accepted as proof of compliance with the Specification but the Engineer reserves the right to carry out further tests in an approved Accredited Laboratory.

The Engineer may without tests being made order any bags of cement a portion of the contents of which has hardened or which appear to be defective in any way to be removed from the site at the Contractor's expense.

The compression test and not the tensile test shall be used to determine the strength of cement.

The cement shall be fresh and delivered in unbroken bags of the manufacturer or in properly designed bulk containers. All cement delivered in bags shall be stored in a weatherproof dry shed with raised boarded floor of at least 300mm above natural ground level.

Cement delivered in bulk tankers shall be stored in properly designed bulk storage tanks. Cement of different manufacture and types shall be kept separate at all times and shall not be used in the same mix.

Concrete of different brands of cement shall not be mixed during casting. The Engineer shall be informed in writing of any change in the source of supply.

Consignments of cement shall be used in the order of delivery and any caked or lumpy cement shall not be used. All cement unfit for use shall be removed from site immediately.

Each consignment shall be identifiable and the Contractor shall supply without cost to the Engineer a copy of Manufacturer's Test Certificate for each consignment.

The age of the cement at the time of delivery to the site shall not be more than 2 months and the cement shall be used within 3 months of delivery.

4.3 Aggregates

Materials used as aggregate shall be obtained from sources known to produce aggregate satisfactory for concrete work.

Materials used as aggregate shall be chemically inert strong, hard, durable, of limited porosity and free from adhering coatings, clay lumps, coal or coal residue and organic or other impurities that may cause corrosion of the reinforcement or impair the strength or durability of the concrete.

Fine aggregates shall be natural sand or crushed gravel sand or crushed stone sand complying with SS EN 12620 or equivalent codes.

Coarse aggregates shall be crushed or uncrushed gravel, crushed or uncrushed stone complying with SS EN 12620 or equivalent codes.

Coarse aggregates shall not exceed 40mm in size for lean and mass concrete and 20mm for reinforced concrete, unless otherwise specified.

All-in aggregates as specified in SS EN 12620 or equivalent code shall not be used. Aggregate used shall not be alkali reactive.

Aggregates delivered to site shall be stored on clean surfaces in separate stockpiles or bunkers to prevent intermingling of different materials and contamination.

The Engineer reserves the right to draw samples of aggregates from any stockpile at any time and have them tested in accordance with BS 812 for compliance with SS EN 12620 or equivalent codes. Aggregates failing to comply with SS EN 12620 or equivalent codes shall be rejected.

Aggregate whose grading is found to fall outside the approved zone shall be rejected.

The Engineer shall be informed in writing of any change in the source of supply of these aggregates.

4.4 Water

Potable water supplied by the Public Utilities Board shall be used for mixing concrete, mortar and grout. For cleaning of moulds and formwork, reinforcement and curing of concrete, water other than potable water may be used if approved by the Engineer. The use of seawater shall be prohibited.

Contractor shall arrange for tests of the water to be carried out in accordance with EN1008.

4.5 Admixtures

When admixture is proposed to be used in concrete mix, furnish information concerning the admixture, admixture acceptance tests and additional information as stipulated in SS EN 934 and SS EN 480 or equivalent codes for approval by the Engineer prior to the use of admixture.

Admixture used shall comply with SS EN 934 and SS EN 480 or relevant Eurocode Standards or equivalent local codes. Requests for permission to use

other types of admixtures not covered by these standards may only be considered when accompanied by full supporting information. Admixture to be used shall be approved by the Engineer.

The Contractor shall provide the Engineer every facility for inspection and sampling of the admixture(s) at least once every three-month of active concreting period. The sample(s) shall be kept for reference and shall be sent, if requested by the Engineer, for admixture uniformity tests in accordance with SS EN 934 and SS EN 480 or the relevant Local codes.

5.0 Concrete Mixes

5.1 The requirements for materials and concrete mixes be in accordance with SS EN 206 and SS EN 544 or equivalent local code.

5.2 Prescribed Mix

Unless otherwise stated, concrete mix shall be designed mix. However, prescribed mix may be used provided that:

- The work is of minor nature or involving a small quantity of concrete:
- Approval is given by Engineer.
- The strength of the concrete is still the responsibility of the contractor.

Concrete Grade	Workability Limits to slump that may be expected	Medium 25-75	High 75-125
C25/30	Cement (kg)	380	410
	Total Aggregate (kg)	1725	1670
	Fine Aggregate *(%by weight of total aggregate)	47	47
	Water (Liter)	210	225

5.3 Designed Mix

For designed mixes, the Contractor shall submit a mix design report covering all concrete mixes to the Engineer for approval. The designed mix shall comply with the following requirements:

Concrete Grade	C15/ 20	C20/ 25	C25/ 30	C28/ 35	C32/ 40	C35/ 45	C40/ 50
Min. Cement Content (kg/m ³)	205	260	275	300	325	350	50
Max. Cement Content (kg/m ³)			550	550	550	550	400
Max. <u>Fine Aggregate</u> (%)			50	50	50	50	550
Total Aggregate							
Max. Water/Cement Ratio	0.70	0.60	0.55	0.50	0.45	0.40	0.40

The minimum cement content shall be increased by 20 and 40 kg/m³ respectively for concrete with maximum aggregate size of 14 and 10mm.

Mix design shall be based on a target mean strength not less than the specified characteristic strength plus 8 N/mm².

In all cases, it is the responsibility of the Contractor to ensure that the designed mix shall be durable, workable and comply with the strength grade requirement as specified in the Drawings.

Design of mix shall ensure no segregation, no excessive bleeding and ability to achieve specified surface finish. Segregation and bleeding shall be determined by visual inspection.

5.4 Trial Mix

The designed mix with a designed maximum consistency as measured by slump or flow shall be verified by preparation of trial mixes witnessed by the Engineer representative. The mix shall be judged for suitable workability as well as strength. A minimum of 12 cubes from three batches shall be taken, six of which shall be tested at 7 days and the other six at 28 days.

The mix may be considered suitable if;

- a. The designed maximum consistency is achieved in accordance with EN 12350, EN 12390, EN 12504 and BS 1881 or equivalent codes if a flow test is specified.
- b. The average strength of the six cubes tested at 28 days exceeds the target mean strength.

No concrete shall be placed in the work until the results of the trial mix are approved by the Engineer.

All the trial mixes and cube tests shall be carried out at the expense of the Contractor. Trial mixes may not be required if concrete is supplied by ready-mix concrete suppliers approved by the Engineer.

5.5 Mix Adjustment

Once the designed mix is approved by the Engineer, the contractor shall not be allowed to change the mix without prior approval of the Engineer.

An increase or reduction of cement content by 20 kg/m³ and above, or % fine aggregate by 5% and above, or any changes in the type or dosage of admixture or consistency is considered a change in mix design. This does not constitute a tolerance allowance on the mix design used in relation to the specified requirements.

The Engineer reserves the right to instruct the Contractor to modify and improve the mix design from time to time during the progress of work as and when he considers necessary to meet the requirements. This shall not entitle the Contractor to any extra payment.

6.0 Batching

Cement shall be measured by weight or batched in whole bags if so packed and delivered. Fine and coarse aggregates shall be measured by weight separately. Water shall be measured by weight and due allowance shall be made for the amount of water present in the mass of fine aggregate by deducting this amount from the weight of water to be added during the mixing of concrete. Solid admixtures shall be measured by weight, while liquid or paste admixtures may be measured by weight or volume.

The accuracy of the measuring equipment shall be within $\pm 3\%$ of the quantity of cement, water or total aggregate being measured, and with $\pm 5\%$ of the quantity of admixture added before commencement of work and subsequently whenever so directed by the Engineer, the measuring equipment shall be calibrated with known weights and volume measures.

7.0 Mixing Concrete

Concrete shall be mixed in a batch type mechanical mixer. The quantity of constituent material in each batch shall not exceed the rated capacity of the mixer. The constituent material shall be thoroughly and continually mixed at the mixing speed specified by the mixer manufacturer until the resulting concrete is uniform in colour and consistency.

The mixing time shall be as follows:

- a. Pan Mixer : Minimum 30 seconds
- b. Truck Mixer or Drum Mixer : Minimum 1.5 minutes for 1m^3 concrete plus $\frac{1}{4}$ minute for additional 1m^3

The mixing time shall be reckoned from the time when all the solid materials are in the mixer. No portion of the time required for discharging shall be considered as part of the mixing time.

The first batch of concrete fed into the mixer shall have an increase of an appropriate amount of dry mortar similar in proportion to that of the concrete to be mixed and of mixing water for coating the mixer.

The mixer shall be emptied completely before being refilled. Mixers which have been out of use for more than 30 minutes should be thoroughly cleaned before any fresh concrete is mixed. The mixer shall be free from hardened concrete. Worn-off blades or paddles shall be replaced.

No concrete shall be mixed by hand except with the specific approval of the Engineer for a small amount. Where such approval is granted, the Contractor shall at his own expense increase the proportion of cement by 10%. Mixing shall be continued until the concrete is of uniform colour and consistency.

8.0 Workability and Consistency

The workability shall be such that the concrete can be well compacted, worked into the corners of the moulds and around the reinforcements, gives the specified surface finish and attain the specified strength. The mix design shall ensure that this workability is obtainable.

On each day of production, workability shall be assessed at the discharge point of the mixer by means of the slump test in accordance with SS 12350, EN 12390 and EN 12504 or equivalent codes. If the measured slump deviates from the agreed value by more than $\pm 25\text{mm}$ or one third of the required value whichever is greater, the concrete shall be rejected.

As an alternative, flow test to BS 1881 : Part 105 or equivalent codes or other means of assessing workability may be considered subject to approval by the Engineer.

No water shall be added to the concrete subsequent to the completion of the mixing.

The required workability for any part of the work shall be determined by the Contractor, subject to the agreement of the Engineer. It shall be reviewed during the progress of the work and adjustments shall be made as necessary subject to the agreement of the Engineer.

9.0 Quality of Concrete Works

9.1 General

Concrete shall be tested for compliance in strength and durability requirements.

The compressive strength of concrete shall be determined by crushing tests of 150mm cubes. The quality and consistency of the concrete shall be tested as directed by the Engineer in accordance with SS 12350, EN 12390 and EN 12504 or equivalent codes. The Contractor shall provide for slump test or flow test, compressive strength test, core test and any other concrete test as and when required by the Engineer.

9.2 Sampling and Testing

All sampling, curing and testing of cubes shall be carried out in accordance with SS12350, EN 12390 and EN 12504 or equivalent codes. A sample shall be taken from a randomly selected batch of concrete chosen by the Engineer.

On each concreting day, at least one sample shall be taken from a batch. A sample shall be a quantity of concrete, consisting of a number of standard scoopfuls, taken from a batch of concrete whose properties shall be determined. Two cubes shall be cast per sample per age of testing unless specified otherwise. The sample shall be taken at the point of discharge from the mixer of a wet batching plant.

At least six cubes shall be cast from each sample. Two cubes shall be tested at an early age of 3 days for precast concrete. The remaining four cubes shall be tested at 7 days and 28 days for compliance purpose.

On completion of the curing, the specimens shall be tested and the mean of the two results shall be taken as the test result. When the difference between the two results divided by their mean exceeds 15%, the test result shall be deemed invalid.

For precast concrete, additional cubes shall be taken from the same sample to determine whether adequate strength is achieved in order to meet design strength for demolding, handling and delivery. These additional cubes shall be cured by the same methods as those used for the precast units. Testing facilities approved by the Engineer shall be provided at precast plant for testing of these additional cubes.

In addition to the minimum number of samples stated above, the Engineer may, at his absolute discretion, instruct the Contractor to take extra samples from whichever batch as he may choose.

The Contractor shall provide suitable equipment for the measurement of concrete cover to reinforcement and in-situ strength of the concrete. The Contractor shall carry out measurements as directed by the Engineer. The Contractor shall analyze statistically all measurements. The concrete cover shall comply with the requirements of SS EN 1992 or equivalent codes. The in-situ strength of the concrete shall be assessed in accordance with EN 13791 or equivalent codes where applicable. The strength shall comply with the specified grade, taking into account the variability with in-situ strength.

The cost for sampling, making, curing, forwarding the cubes for testing and measuring of concrete cover and in-situ strength shall be borne by the Contractor. The cubes for testing shall be sent to an Accredited testing laboratory approved by the Engineer. Testing fees will be borne by the Contractor. Should the Engineer so requires, the Contractor shall provide facilities for making specimens for Flexural Strength Tests in accordance with EN 12350, EN 12390 and EN 12504 or equivalent codes.

9.3 Compliance of Strength Requirement

For precast concrete, the 3 days strength shall be obtained solely for quality control purpose. Early strength requirements to meet design strength for demolding, handling and delivery shall be specified by the approved manufacturer of the precast components and approved by the Engineer. Such strength shall be determined from the additional cubes. The Engineer shall have the absolute right to reject any precast components should the early strength fail to meet the design strength requirements.

Compliance with the specified characteristic strength shall be judged by tests made on cubes at the age of 7 and 28 days. Compliance may be assumed if both the following conditions are satisfied:

- (i) The average strength of the sample and the preceding three samples or the average of the first four samples of the production is greater than or equal to the specified characteristic strength plus half a fixed current margin of 6 N/mm². Thus, the minimum average strength shall be:

Grade	C25/30	C28/35	C32/40	C35/45	C40/50
Minimum Average 7 Days	24	28	32	36	40
Strength (N/mm ²) in 28 Days	33	38	43	48	53

(ii) The sample strength is at least the level specified below:

Grade	C25/30	C28/35	C32/40	C35/45	C40/50
Minimum AIndividual 7 Days	18	22	27	30	34
Strength (N/mm ²) in 28 Days	27	32	37	42	47

All test cubes taken shall be properly stored and the Contractor shall ensure that all test cubes are tested at their respective age. Test cube that are lost shall be deemed to have failed.

9.4 Action to be Taken in the Event of Non-Compliance

If the sample strength results do not comply with either or both of the above conditions the Contractor shall redesign his mix based on an increased target mean strength. The section of work represented by the samples which fail the test shall be rejected and removed. The Engineer may direct that other measures be taken to make the works secure. The Engineer may instruct that additional tests be carried out on the hardened concrete.

The affected concrete work may be stopped and not continued until so directed by the Engineer. No extension to the contract period shall be allowed for such stoppage of the work and remedial works.

All works instructed under this sub-clause shall be at the Contractor's expense.

9.5 Cube Tested at 7 Days

When the 7-days strength is less than that stipulated in Clause 17.3, no more concrete shall be placed on the suspect concrete until the 28-days strength is found to comply.

Any additional expense and delay shall be borne solely by the Contractor.

10.0 Testing of Hardened Concrete

In the event of non-compliance, the quality of concrete in question shall be interpreted in accordance with SS EN 206 or equivalent codes.

For 7 days non-compliance, prolonged propping shall be provided until the 28 days strength becomes available.

For 28 days non-compliance, the quality of the sub- standard concrete shall be estimated by carrying out tests on hardened concrete in the structure. These may include non- destructive methods, the taking of core samples, load tests or combination of several methods as follows:

Grade	Average Compressive Strength	Test on Structure
C25/30	27.0 to below 33.0	Non-destructive method and/or Core test
C28/35	32.0 to below 38.0	
C32/40	37.0 to below 43.0	
C35/45	42.0 to below 48.0	
C40/50	47.0 to below 53.0	
C25/30	Below 27.0	Non-destructive method and Core test and/or Load test
C28/35	Below 32.0	
C32/40	Below 37.0	
C35/45	Below 42.0	
C40/50	Below 47.0	

The test shall be carried out by an accredited laboratory in accordance with SS EN 1992 and assessed in accordance with EN 13791 or equivalent codes. The results of any such tests shall not nullify the establishment of non-compliance provided that the latter are based on valid cube tests results. Notwithstanding the outcome of such tests, the Contractor shall bear all the costs and expenses incurred for such tests.

10.1 Core Test

The Engineer reserves the right to direct that samples be cored from any portion of works for the purpose of examination and testing.

The cutting equipment and method of coring to obtain specimens shall be approved by the Engineer

The specimens shall be dealt in accordance with BS 1881. Prior to preparation for testing, the specimens shall be made available for examination by the Engineer.

The cored portion of the works shall be made good to the satisfaction of the Engineer.

10.2 Check Test

Consequent upon work cube test failure, the Engineer may instruct that other appropriate tests be carried out in accordance with SS EN 1992 or equivalent codes on any portion of the works for the purpose of assessing the strength. The Contractor shall bear all the cost for such test.

11.0 Structural Tests

11.1 General

The Engineer may instruct the Contractor to make a load test on the works or any part thereof if in the Engineer's opinion such a test is necessary.

The Contractor shall bear all the costs and expenses incurred for such test.

11.2 Load Test

Precast Units

Testing of the precast units shall be carried out as and when directed by the Engineer in accordance with SS EN 1992 or equivalent codes. Samples of the precast unit for the tests shall be provided and delivered by the Contractor at his cost to an approved laboratory.

Where more than one sample from any sample batch of two or more similar units fail to satisfy this Specification, then the entire batch of units represented by such samples shall be liable for rejection by the Engineer.

12.0 Rectification of Defective Works

The action to be taken in respect of the concrete which fails to meet either of the requirements in Clause 10 'Quality of Concrete Works' and the assessment specified in Clause 11 'Testing of Hardened Concrete' shall be determined by the Engineer. They may range from:

12.1 Qualified acceptance in less severe cases. The Contractor shall provide protective coating against carbonation and other aggressive environment if the affected concrete can be certified to be structurally sound by a Professional Engineer engaged by the Contractor at its own costs and expenses; or

12.2 If the average of three core tests falls below the strength level specified in Clause 17.3(ii), or in the event of failure of a load test.

The Contractor shall carry out at his own costs and expenses any necessary approved remedial measures to the satisfaction of the Engineer. Remedial measures shall be designed, supervised and certified by a Professional Engineer engaged by the Contractor at its own costs and expenses.

13.0 Steel Reinforcement

13.1 Description

Unless otherwise shown on the Drawings, reinforcement for concrete shall consist of plain mild steel and high yield reinforcement bars complying with SS 560 and welded steel fabric complying with SS 561. Their respective specified characteristic strength shall not be less than 250 N/mm², 500 N/mm² and 500 N/mm².

Reinforcement shall be free from pitting, loose rust, mill scale, paint, oil, grease, adhering earth or any other material that may impair the bond between the concrete and the reinforcement or that may cause corrosion of the reinforcement or disintegration of the concrete.

13.2 Testing

The Contractor shall provide the Engineer every facility for inspection and sampling before any cutting, bending or other fabrication is undertaken. If deemed necessary, the Engineer may cause samples to be taken from reinforcement already fixed in position.

Steel reinforcement shall be tested by an Accredited Laboratory approved by the Engineer. Reinforcement not complying with the appropriate specifications shall be removed and replaced at the Contractor's own costs and expenses.

The reinforcement shall be tested in accordance with the relevant Singapore Standard. Certificates of origin and manufacturer's test certificates shall be produced. Selection of test pieces shall be made by the Engineer. The Contractor shall bear all costs involved in making these tests. The Contractor shall have no claim for bars mutilated in obtaining the test pieces.

The Contractor shall furnish, for reinforcement supplied by him, manufacturers' certificates and these shall be submitted for approval by the Engineer before any materials is brought on to the site.

The Contractor shall also submit test record issued by approved laboratory giving the test results of the mechanical properties including tensile strength, yield point and elongation.

13.3 Galvanized Steel

Galvanized mild steel and galvanized high yield steel reinforcement bars as indicated in the Drawings shall be hot-dip galvanized to EN ISO 1460 and EN ISO 1461 or equivalent codes. The Contractor shall obtain galvanizing service or galvanized steel bars only from approved galvanizers or suppliers.

The coating weight of zinc shall conform with EN ISO 1460 and EN ISO 1461 or equivalent codes. The coating shall be continuous, smooth and free from flux stains. Should the coating appearance of any galvanized reinforcement bar be found otherwise, that bar shall not be used as a galvanized reinforcement bar. Galvanizing shall be followed by chromating as a post treatment. Store all galvanized reinforcement bars in a dry and well

ventilated place to prevent the formation of white rust on these bars. Small areas of galvanized coating damaged by welding, cutting or by excessively rough treatment shall be made good by the application of at least two coats of good quality zinc-rich paint with minimum 94% of zinc dust, expressed as a percentage by weight of the solid content of the paint.

Upon the instruction of the Engineer, the Contractor shall send at least two samples of the galvanized reinforcement bars to be tested by a competent authority to determine the coating weight of these bars. The coating weight shall be determined by stripping tests in accordance with EN ISO 1460 and EN ISO 1461 or equivalent codes. Certificates of such tests shall be submitted to the Engineer for verification.

14.0 Fabrication of Reinforcement

All reinforcement shall be accurately cut and bent to the dimensions and shapes shown in the Drawings, and in accordance with EN ISO 3766 or equivalent codes.

Bends shall be cold-formed by machine or by other approved means which produce a gradual and even bending action. Reinforcement incorrectly bent shall not be used unless the method for straightening and rebending will not cause any cracking, and approval for rebending is given by the Engineer.

Reinforcement shall be securely tied with steel wires. The projecting ends of the steel wires shall be directed away from the formwork so as not to encroach into the concrete cover.

Reinforcement shall be adequately supported by approved spacers and chairs to maintain the specified concrete cover.

The reinforcement shall be thoroughly wire-brushed and cleaned to remove loose rust mill scale paint, oil grease adhering earth or any other material that may impair the bond between the concrete and the reinforcement or that may cause corrosion of the reinforcement or disintegration of the concrete before placing the bars into position.

Minimum cover to reinforcement shall be as specified in the drawings.

15.0 Schedule of Construction Tests for Concrete Works

The contractor shall allow and conduct all construction tests as follows:

Such test shall be carried out by an accredited laboratory.

Test To Be Carried Out	Standard To Be Used or Equivalent Local Code	Minimum Frequency of Testing / The Numbers of Testing Required
1. Cement	SS EN 197-1	To obtain from concrete supplier.
2. Aggregate	BS 812 For Compliance with SS EN 12620	
3. Water	EN 1008	
4. Concrete Cubes	EN 12350 EN12390 EN12504 BS 1881	As specified
5. Steel Reinforcement a. Tensile Test b. Bend Test and c. Rebend Test	SS 569 SS 561	1 sample of 3 Test Pieces per size per consignment

16.0 Document Control

16.1 Approval: GLOBAL_FAC_GFTT_APPR

16.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

16.3 Notification:

All Micron Team Members at all Sites

16.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

16.5 Review

Biennially (two years)

17.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	09/22/2022